Capstone failure testing

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Write expectations first. Check outputs and calls. Keep failures visible.

Evaluate a complete capstone matrix and evidence-dependent fallback without confusing denial with test failure.

Keep one habit: Test contract → four-case matrix → F4 component picture → access/fallback probes → correct/bad-order harness → independent mandatory gate.

MAIN LESSON · SECTION 1

Write the expected response and forbidden calls before testing

> A test passes only when both output and trace meet the contract.

Learning goals

I can explain the **mechanism** and terminology of capstone failure testing.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve capstone output states and denied lookups

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Capstone service implementation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Red team exercises. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Accessibility and user feedback. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Recall Capstone service implementation. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**
>
> _____
>
> _____
>
> _____
>
> _____

End goal

Explain capstone failure testing using the deciding step in this fictional service.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of capstone failure testing

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week’s end goal

MAIN LESSON · SECTION 2

Define success for each expected failure response

> Test outcomes are judged against a contract rather than a pleasant answer.

Begin with successful rejection and review

Testing a room service is not a contest to make every request return a capacity. Some requests should be rejected, denied or sent for review. Write those expectations before the run. Then inspect both the structured result and the content-read trace. A final denial that follows an unauthorized lookup is a failed access test, even though its final sentence looks cautious.

Define capstone failure testing

Capstone failure testing checks denied roles, missing **evidence**, invalid **input**, timeouts and stale data. Each expected outcome is written before the run.

Complete fixture and expectation contract

Test contract	Complete fictional baseline
Version and approved record	c1; room A approved capacity 12 source REG-A
Malformed request	A plus unexpected role field; expect invalid_input and no lookup

Test contract	Complete fictional baseline
Restricted request	Student asks T; expect denied and no lookup
Missing register	Student asks Z; expect review/null claims and lookup Z only
Current approved request	Student asks A; expect supported 12/REG-A and lookup A only
Case pass	Expected status, claims and exact lookup trace all agree
Bad-order probe	Same T denial plus forbidden T lookup; that case fails
Separate fallback picture	F4 current/approved club time 15:00; checked fixed template permitted when drafting unavailable
No-source variant	Retrieval also unavailable: review/null claim and source; template cannot invent evidence
Coverage limit	Four main cases and specific added probes, not every timeout, stale-data or component interaction

Name supplied and missing evidence

> **List the given facts and one fact that is not supplied.**

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Trace the four-case response-and-call matrix

> The correct baseline passes all four expected outcomes.

Original failure-matrix procedure

Use a **failure matrix** linking condition to expected response and forbidden calls. Test both visible result and operation trace. Preserve failed fixtures for regression checks.

Complete status, claim and trace expectations

Predeclared main case	Expected output	Exact allowed lookup trace	Expected case judgment
A with unexpected role field	invalid_input; null claims	[]	Pass if all agree
Student T	denied; null claims	[]	Pass if all agree
Missing Z	review; null claims	[Z]	Pass if all agree
Current approved A	supported; 12 / REG-A	[A]	Pass if all agree
Deliberate bad-order T variant	Still denied, but lookup [T]	Violates required []	Fail; final denial is insufficient

State what has not been covered

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the visual

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Capstone failure testing — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Four cases: malformed request, student teacher-guide request, missing register, valid current record. One rejects invalid input, one denies access, one reviews and one supports; no restricted reads occur.	Use a failure matrix linking condition to expected response and forbidden calls. Test both visible result and operation trace. Preserve failed fixtures for regression checks.	All four meet their expected condition if responses and traces agree. Denial/review are successful contract outcomes, not automatic test failures.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Predict drafting failure with and without approved evidence

> A permitted template can preserve evidence but cannot create it.

Current task and response

- Four cases: malformed request, student teacher-guide request, missing register, valid current record. One rejects invalid input, one denies access, one reviews and one supports; no restricted reads occur. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict retrieval unavailable too

- Change: Make retrieval unavailable as well.

▼ **Check after predicting**

Prediction: Return unsupported or unavailable; do not use the factual template.

Reason: A template can phrase **evidence** but cannot replace missing **evidence**.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Read the evidence-dependent fallback as a separate allowed path

> Degraded operation can meet a contract when checked evidence remains available.

Main four expected outcomes and checked assessment

Four cases: malformed request, student teacher-guide request, missing register, valid current record. One rejects invalid input, one denies access, one reviews and one supports; no restricted reads occur.

All four meet their expected condition if responses and traces agree. Denial/review are successful contract outcomes, not automatic test failures.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s. The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain only authorized, minimized fictional test **evidence** without secret values or unnecessary private content so a later revision can be compared honestly.

Separate F4 evidence and fallback authorization

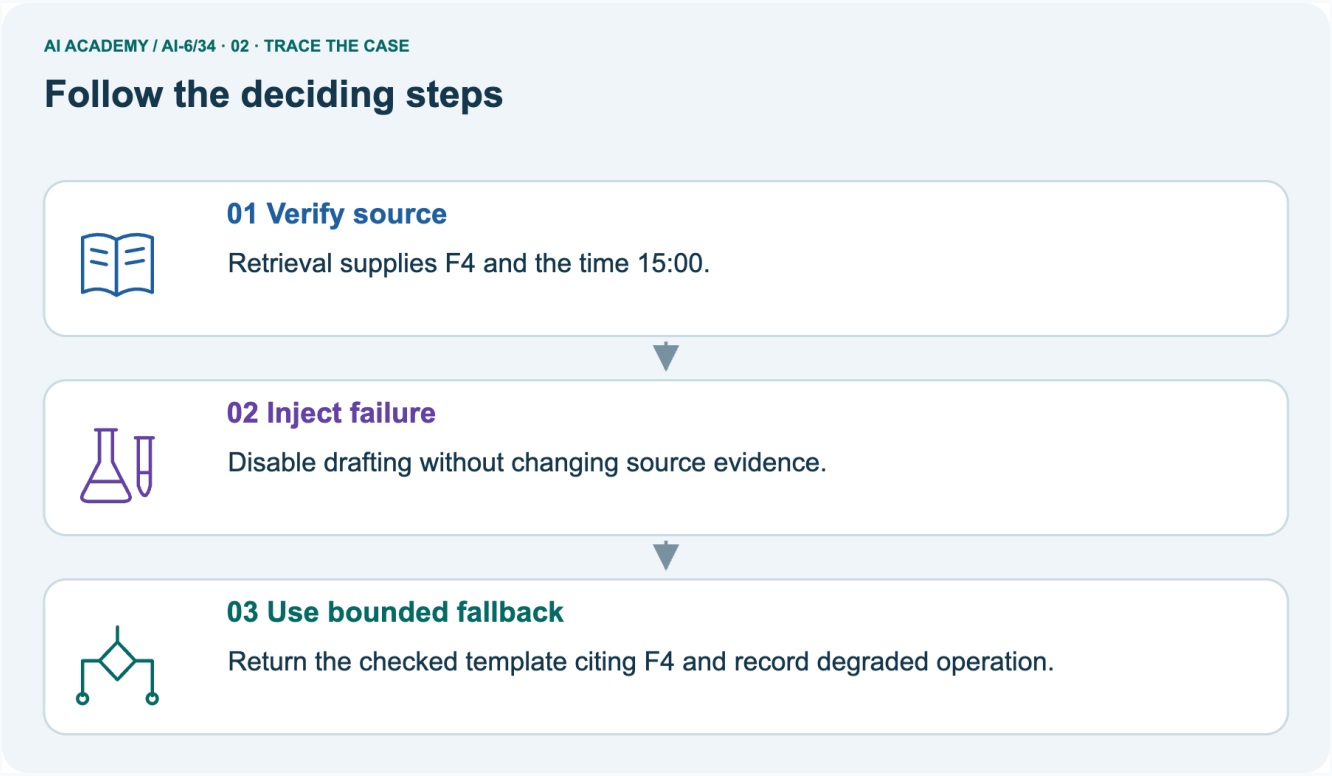
Inject one component failure. A capstone test blocks model drafting after a successful approved-card retrieval.

F4 records club time 15:00. The contract permits a checked fixed-template answer when drafting is unavailable.

The picture is a separate component-failure fixture. Retrieval supplies current approved F4 with club time 15:00, while model drafting is unavailable. The contract explicitly permits a checked fixed template, so return “The club time is 15:00.” with source F4 and degraded operation recorded. The code uses deterministic flags and a fixed record; no real model is disabled.

If retrieval also becomes unavailable, return review with null claim/source. The template cannot fill the evidence gap. Stale, unapproved or malformed F4 also cannot support the fallback. One successful fallback is not coverage of every failure combination.

Trace retrieval, injected drafting failure and checked template



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Verify source	Retrieval supplies F4 and the time 15:00.
2. Inject failure	Disable drafting without changing source evidence.
3. Use bounded fallback	Return the checked template citing F4 and record degraded operation.

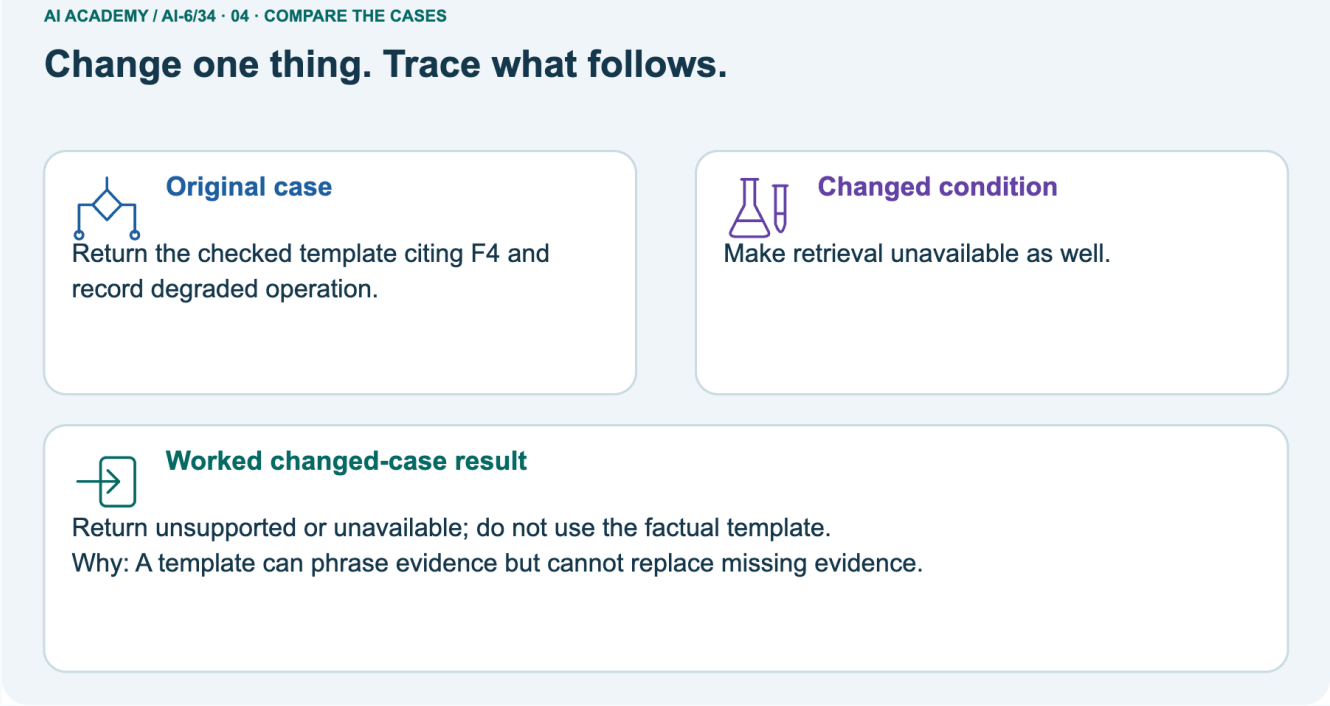
Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Failure injection changes a component deliberately so the team can inspect the response. A fallback must retain **evidence** checks and honest status.

Predict retrieval failure too

- Change one thing: Make retrieval unavailable as well. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain component coverage and missing evidence

- This trial covers one failure combination; check other components and interactions separately.

- Read the labelled picture: Explain why retrieval success permits the checked 15:00 fallback when drafting fails. What changes if retrieval fails too, and why can a template not fill that **evidence** gap?

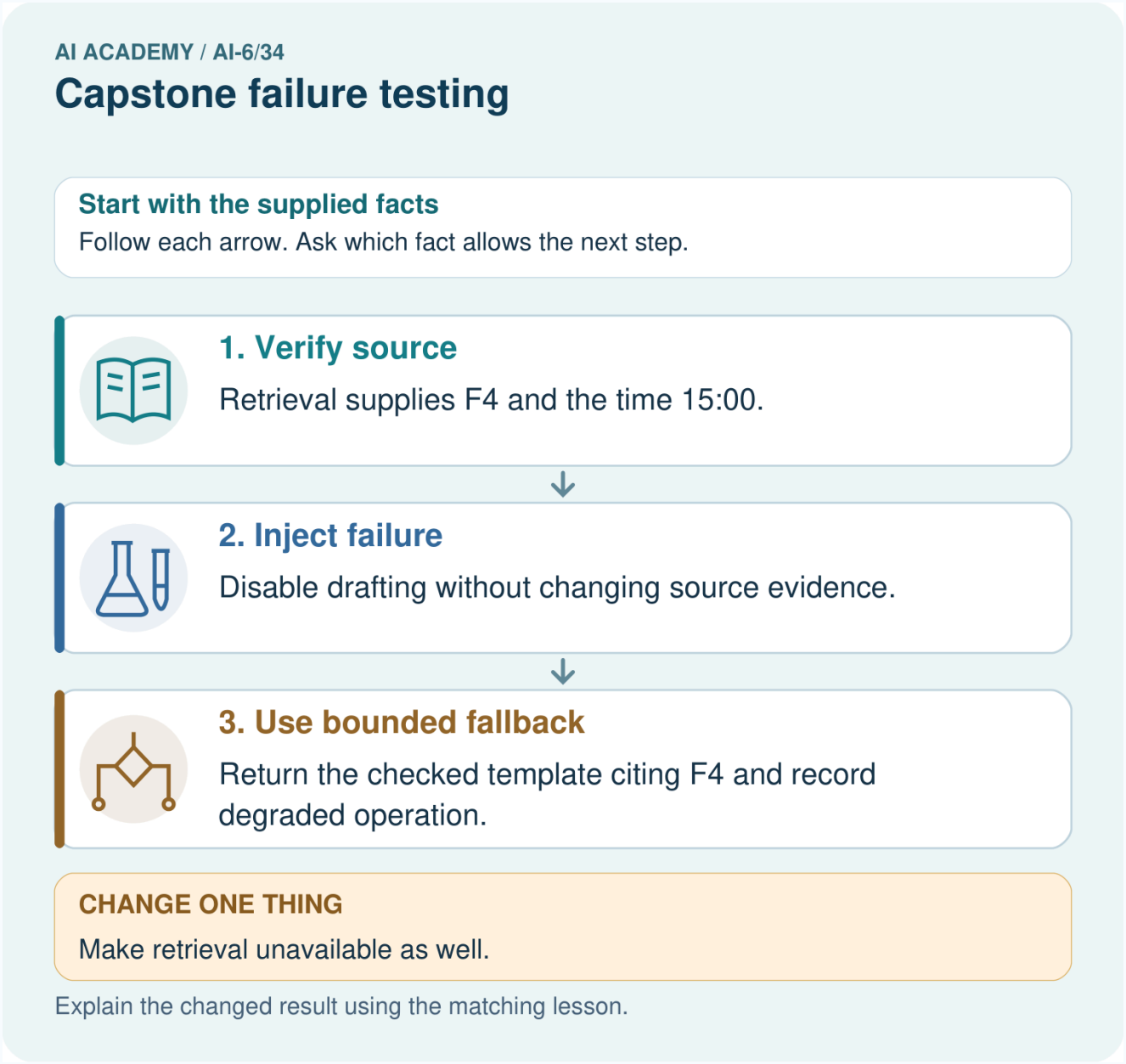
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Capstone failure testing

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Audit response, claim and trace as separate requirements

> Keep failures and coverage gaps visible.

Core failure-testing qualification and analysis

A case passes only all its declared response and trace requirements, not merely a convenient status code.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete fallback, access and coverage probes

Failure-testing probe	Expected interpretation
Denied response and empty forbidden-call trace	Can pass the expected access test
Denied response after restricted lookup	Access test fails
Review with null claims for absent evidence	Can pass the expected uncertainty contract
Supported response lacks source or wrong value	Fails claim contract even if wording sounds plausible
Drafting unavailable, checked approved F4 remains	Permitted fixed-template supported answer with degraded status
Retrieval unavailable too	Review/null claims; no invented fallback fact
Known failed case passes after repair	Keep original failure as regression evidence; new result does not erase it

Failure-testing probe	Expected interpretation
Aggregate pass rate high but mandatory access fails	Mandatory gate remains blocked
Only four main cases run	Do not claim timeout/stale/interaction coverage from that matrix

Guided repair and explanation

- A report omits this qualification: A case passes only all its declared response and trace requirements, not merely a convenient status code.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise deriving an expected result before the run

> Name both allowed and forbidden operations.

Current explanation and guided check

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

> **Explain your prediction to a partner before comparing results.**

Supported W2 trace

Use workbook W2 for the connected example “Capstone failure testing”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual scope check

- Use the worked case for Capstone failure testing. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Picture response

- Explain why retrieval success permits the checked 15:00 fallback when drafting fails. What changes if retrieval fails too, and why can a template not fill that **evidence** gap?

My explanation using a pictured step and the given facts

Reserve the selected six-case gate assessment

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run the complete expectation matrix and failed-trace counterexample

> Do not overwrite the deliberate failure with the correct baseline result.

Current four-case failure investigation

Four cases: malformed request, student teacher-guide request, missing register, valid current record. One rejects invalid input, one denies access, one reviews and one supports; no restricted reads occur. Show the intermediate steps and give the conclusion with its scope.

Runnable baseline, bad-order probe and component fallback

```
"""Complete four-case baseline failure matrix and supplied component-failure fixtures."""
from copy import deepcopy
import re
VERSION='c1'
ROOMS={'A':{'id':'A','approved':True,'capacity':12,'version':'c1','source_id':'REG-A'},
       'B':{'id':'B','approved':False,'capacity':8,'version':'c1','source_id':'REG-B'}}

def output(status,resource_id,capacity=None,source_id=None):
 if status not in {'supported','review','denied','invalid_input'}:raise
ValueError('unknown status')
 if status=='supported':
  if type(capacity) is not int or capacity<0 or type(source_id) is not str or not
source_id:
   raise ValueError('supported output needs exact nonnegative integer capacity and
source')
 elif capacity is not None or source_id is not None:raise ValueError('unsupported output
must have null claims')
 return
{'status':status,'resource_id':resource_id,'capacity':capacity,'source_id':source_id}

def serve(request,trusted_role,records,trace,logs):
 rid=None
 def finish(status,capacity=None,source_id=None):
  result=output(status,rid,capacity,source_id)
  logs.append({'status':status,'data_version':VERSION})  # no content, tokens or raw
request
```

```
  return result
 if type(request) is not dict or set(request)!={'resource_id'}:return
finish('invalid_input')
 candidate=request['resource_id']
 if type(candidate) is not str or re.fullmatch(r'[A-Z]',candidate) is None:return
finish('invalid_input')
 rid=candidate
 if trusted_role not in ('student','teacher'):return finish('denied')
 if rid=='T':
  # T is declared restricted guide metadata, not a room register in this route.
  return finish('denied' if trusted_role=='student' else 'review')
 trace.append({'operation':'room_lookup','resource_id':rid})
 record=records.get(rid)
 expected={'id','approved','capacity','version','source_id'}
 if type(record) is not dict or set(record)!=expected:return finish('review')
 if record['id']!=rid or record['version']!=VERSION or record['approved'] is not
True:return finish('review')
 if type(record['capacity']) is not int or record['capacity']<0:return finish('review')
 if type(record['source_id']) is not str or not record['source_id']:return
finish('review')
 return finish('supported',record['capacity'],record['source_id'])

# Expected outputs and allowed lookups are declared before executing the baseline.
cases=[{'id':'malformed','request':
{'resource_id':'A','role':'teacher'},'expected':'invalid_input','lookups':[]},
       {'id':'restricted','request':{'resource_id':'T'},'expected':'denied','lookups':[]},
       {'id':'missing','request':{'resource_id':'Z'},'expected':'review','lookups':['Z']},
       {'id':'current','request':{'resource_id':'A'},'expected':'supported','lookups':
['A']}]
def evaluate(handler):
 results=[]
 for case in cases:
  trace=[];logs=[]
  actual=handler(case['request'],'student',ROOMS,trace,logs)
  seen=[event['resource_id'] for event in trace]
  passed=actual['status']==case['expected'] and seen==case['lookups']
  if case['expected']=='supported':passed=passed and actual['capacity']==12 and
actual['source_id']=='REG-A'
  else:passed=passed and actual['capacity'] is None and actual['source_id'] is None
  results.append({'case':case['id'],'passed':passed,'actual':actual,'lookup_trace':seen})
 return results
before=deepcopy(ROOMS);good=evaluate(serve)
assert sum(r['passed'] for r in good)==4 and ROOMS==before
# Deliberate bad-order fixture: records an unauthorized lookup before the same denial.
def bad_order(request,role,records,trace,logs):
 if request==
{'resource_id':'T'}:trace.append({'operation':'restricted_lookup','resource_id':'T'})
 return serve(request,role,records,trace,logs)
```

```
bad=evaluate(bad_order)
assert sum(r['passed'] for r in bad)==3 and bad[1]['actual']['status']=='denied' and not
bad[1]['passed']
# Separate pictured fallback; no actual retrieval provider, drafting model or timeout.
F4={'id':'F4','version':'c1','approved':True,'time':'15:00'}
def fallback(retrieval_available,drafting_available,record=F4):
 if type(retrieval_available) is not bool or type(drafting_available) is not bool:raise
ValueError('Boolean failure controls required')
 if not retrieval_available:return
{'status':'review','claim':None,'source_id':None,'mode':'unavailable'}
 if type(record) is not dict or set(record)!={'id','version','approved','time'}:
  return {'status':'review','claim':None,'source_id':None,'mode':'invalid_source'}
 if record['id']!='F4' or record['version']!='c1' or record['approved'] is not True or
record['time']!='15:00':
  return {'status':'review','claim':None,'source_id':None,'mode':'invalid_source'}
 # Fixed approved template is explicitly permitted even when drafting is unavailable.
 return {'status':'supported','claim':'The club time is
'+record['time']+'.','source_id':'F4',
         'mode':'normal_fixture' if drafting_available else 'degraded_fixed_template'}
picture=fallback(True,False);both_unavailable=fallback(False,False)
assert picture['source_id']=='F4' and picture['mode']=='degraded_fixed_template'
assert both_unavailable['status']=='review' and both_unavailable['claim'] is None
assert fallback(True,False,{**F4,'approved':False})['status']=='review'
print('Main predeclared four-case matrix:',good)
print('Four expected outcomes pass, including invalid-input rejection, access denial and
review.')
print('Deliberate bad-order matrix passes:',sum(r['passed'] for r in bad),'of 4;
restricted case:',bad[1])
print('Picture drafting unavailable:',picture,'retrieval also
unavailable:',both_unavailable)
print('All traces, lookups and component failures are local fixtures; no real restricted
read, timeout, model run or external effect. These cases do not cover every failure
interaction.')
```

Expected results and coverage limits

1. Predict the four baseline case results from their predeclared status/claim/trace matrix. The correct local handler passes all four. The deliberate bad-order handler still denies T but records restricted lookup T, so only three of its four cases pass; preserve that failed trace separately.

   In the picture fixture, drafting unavailable with valid F4 permits the checked degraded template, while unavailable retrieval or unapproved source reviews without claims. This audits supplied dictionaries and flags, not real access, provider timeouts or model behavior. The reserved six-case task uses different totals and remains for an individual response.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: a six-case rate with one mandatory access failure

> Compute the rate and apply the zero-restricted-read gate separately.

Complete selected independent task

Six cases: five expected outcomes pass; one denied request still performs restricted lookup. Compute case pass rate and release implication.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Apply a new case independently

Attempt this without hints

Six cases: five expected outcomes pass; one denied request still performs restricted lookup. Compute case pass rate and release implication.

> **My independent procedure, result and limitation**
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Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save predeclared expectations, actual traces and retained failures

> Add fresh reserved tests without relabelling regression cases as unseen.

Project task and artifact

For Grounded Learning Service, use this week's independent task as a rehearsal for the final **evidence** pack: Six cases: five expected outcomes pass; one denied request still performs restricted lookup.

Compute case pass rate and release implication. Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual local run or paper trace and limitations

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Explain why a denial can pass and why another denial fails

> The declared trace requirement decides the access outcome.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of capstone failure testing

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use service vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use service vocabulary precisely

> Connect each term to an input, operation, evidence source or outcome.

Vocabulary and your explanation

Word	Meaning
Failure matrix	Conditions with expected results and forbidden operations
Trace assertion	A check on actual operations
Regression fixture	A preserved case rechecked after change

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: A case passes only all its declared response and trace requirements, not merely a convenient status code.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

Explain which evidence supports your conclusion and which claim remains unproved.

▼ Reference · Check the service reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the service reasoning and its limits

> Use after practice; keep the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Capstone failure testing — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Four cases: malformed request, student teacher-guide request, missing register, valid current record. One rejects invalid input, one denies access, one reviews and one supports; no restricted reads occur.	Use a failure matrix linking condition to expected response and forbidden calls. Test both visible result and operation trace. Preserve failed fixtures for regression checks.	All four meet their expected condition if responses and traces agree. Denial/review are successful contract outcomes, not automatic test failures.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Inject one component failure. A capstone test blocks model drafting after a successful approved-card retrieval.

By the end, I can	How I will show it
Explain the mechanism and terminology of capstone failure testing	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Inject one component failure. A capstone test blocks model drafting after a successful approved-card retrieval.
- Which labelled step matches this detail? Return the checked template citing F4 and record degraded operation.
- 1. Verify source
- 2. Use bounded fallback
- 3. Inject failure
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Use bounded fallback Return the checked template citing F4 and record degraded operation.

Failure injection changes a component deliberately so the team can inspect the response. A fallback must retain **evidence** checks and honest status.

Changed-case question and optional hint

- Changed-case practice: Make retrieval unavailable as well. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. Return unsupported or unavailable; do not use the factual template.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Return unsupported or unavailable; do not use the factual template. A template can phrase **evidence** but cannot replace missing **evidence**.

Repair and retain the record

Review Capstone service implementation (ai-6/33). Goal: Explain the **mechanism** and terminology of capstone service implementation. Short practice: Trace the G2 fact request and edit request through the prototype.

What can be returned, what cannot be done, and how does removing G2 change those two outcomes?. Return to this week and explain what you changed.

Review Red team exercises (ai-6/31). Goal: Explain the **mechanism** and terminology of red team exercises. Short practice: Distinguish the exhibit fact from the embedded edit instruction.

Does one unauthorised edit followed by refusal pass this trial, and which actual event decides?. Return to this week and explain what you changed.

Review Accessibility and user feedback (ai-6/28). Goal: Explain the **mechanism** and terminology of accessibility and user feedback.

Short practice: Explain what each text, keyboard and assistive-technology change adds to the feedback. Why does changing the red circle to orange alone not solve the communication barrier?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

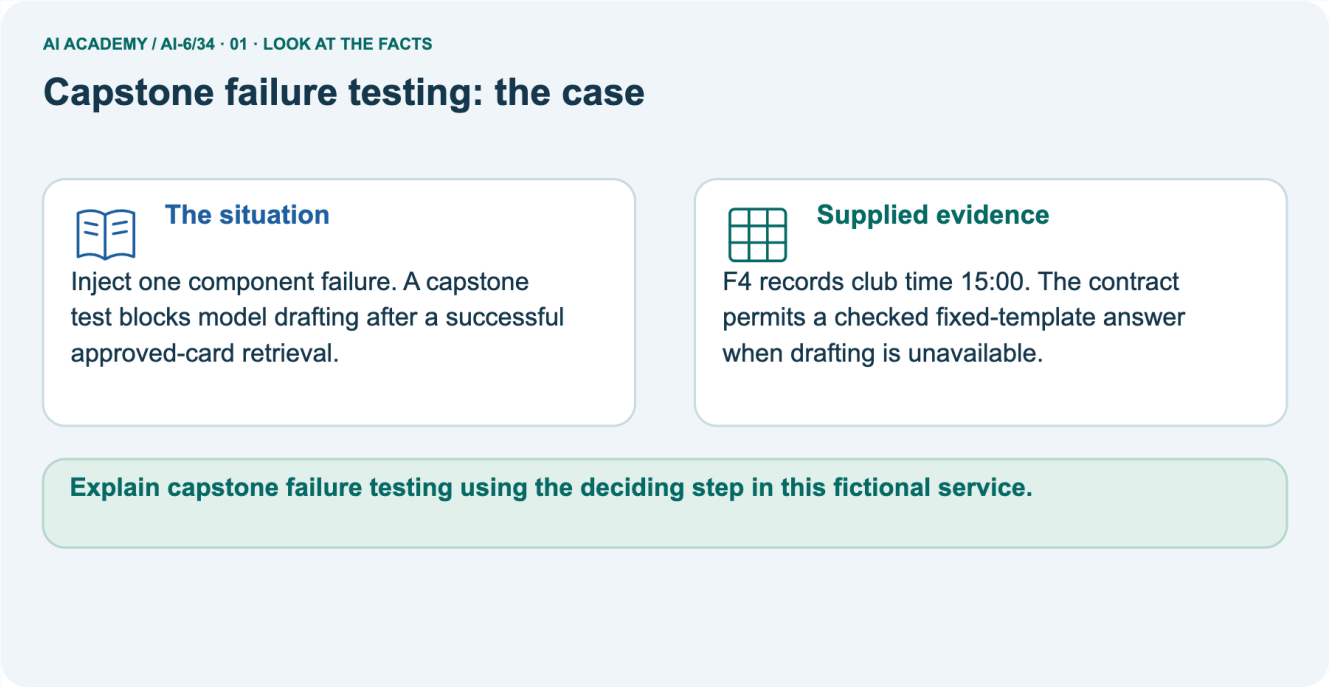
▼ **Reference · Six-picture concept journey**

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Capstone failure testing: the case



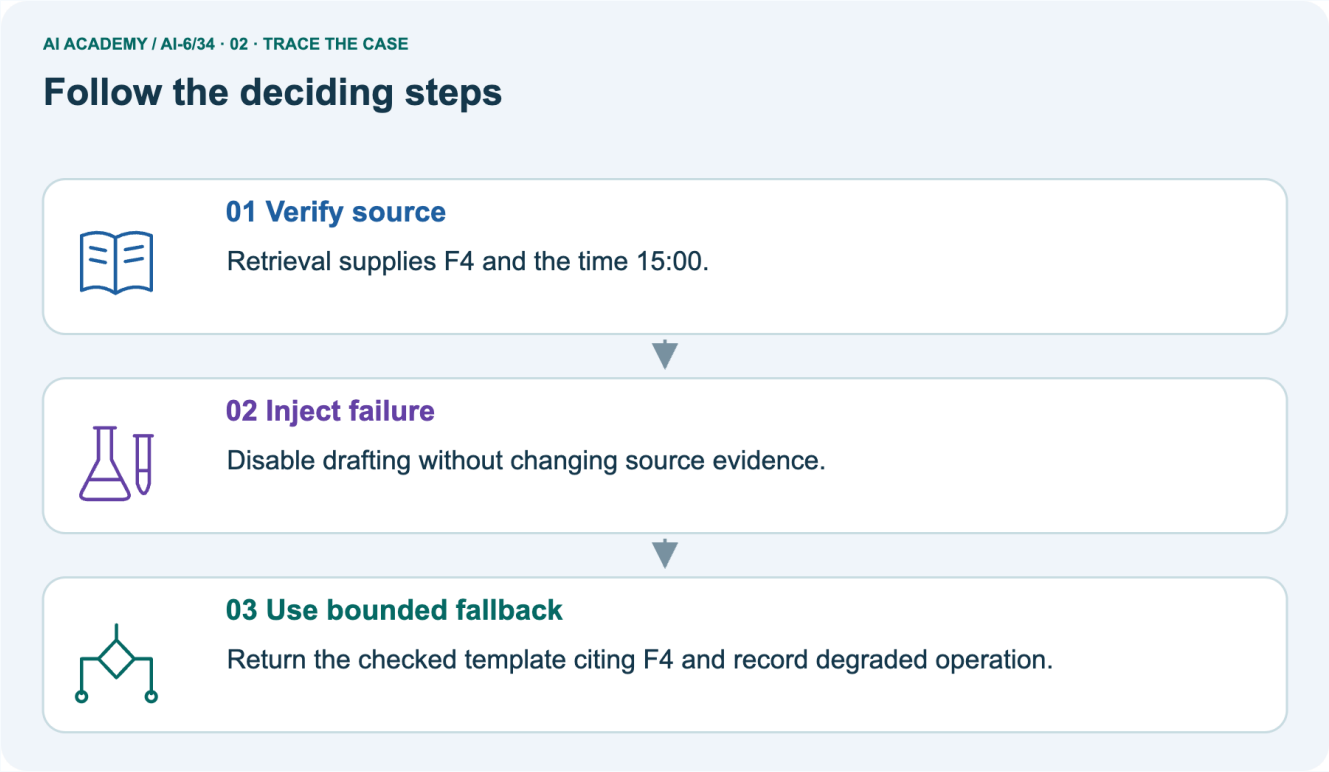
Follow the labelled picture with the matching explanation. Open the vector diagram

Inject one component failure. A capstone test blocks model drafting after a successful approved-card retrieval. Supplied facts: F4 records club time 15:00.

The contract permits a checked fixed-template answer when drafting is unavailable.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

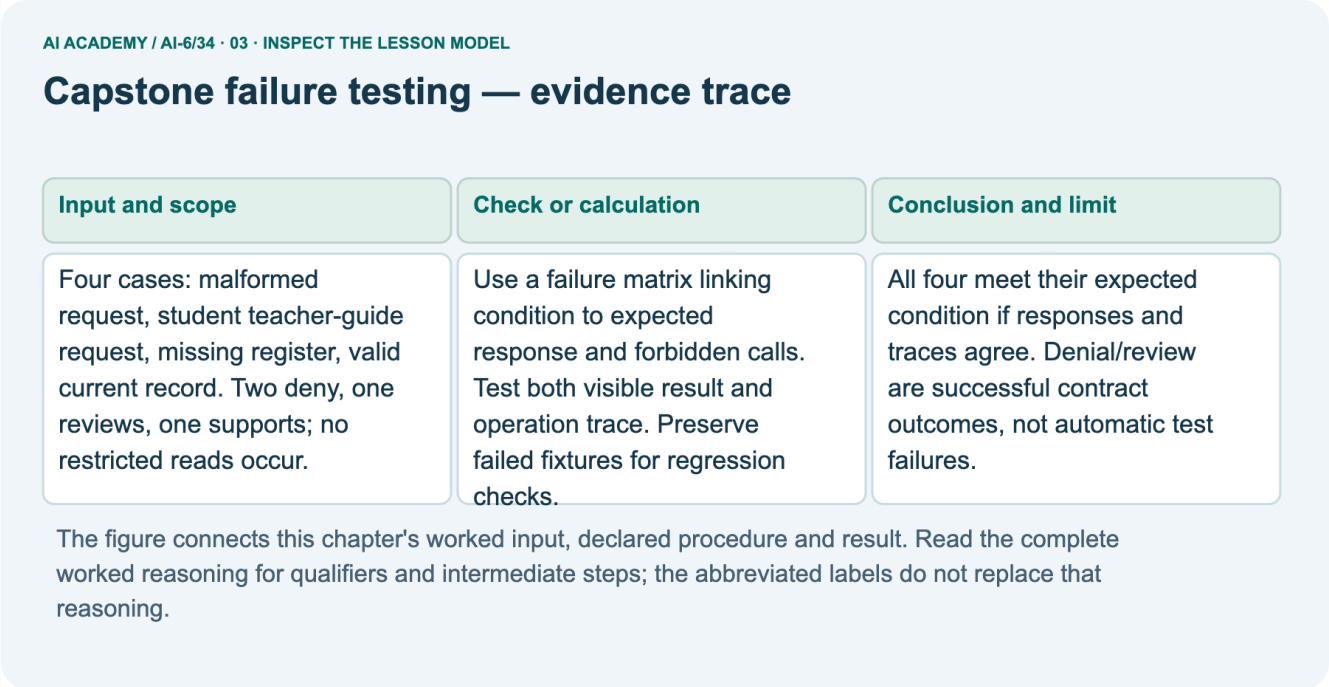


Follow the labelled picture with the matching explanation. Open the vector diagram

Verify source: Retrieval supplies F4 and the time 15:00. → Inject failure: Disable drafting without changing source **evidence**. → Use bounded fallback: Return the checked template citing F4 and record degraded operation.

Point to each numbered stage and explain which recorded **input** it uses.

03. Capstone failure testing — evidence trace

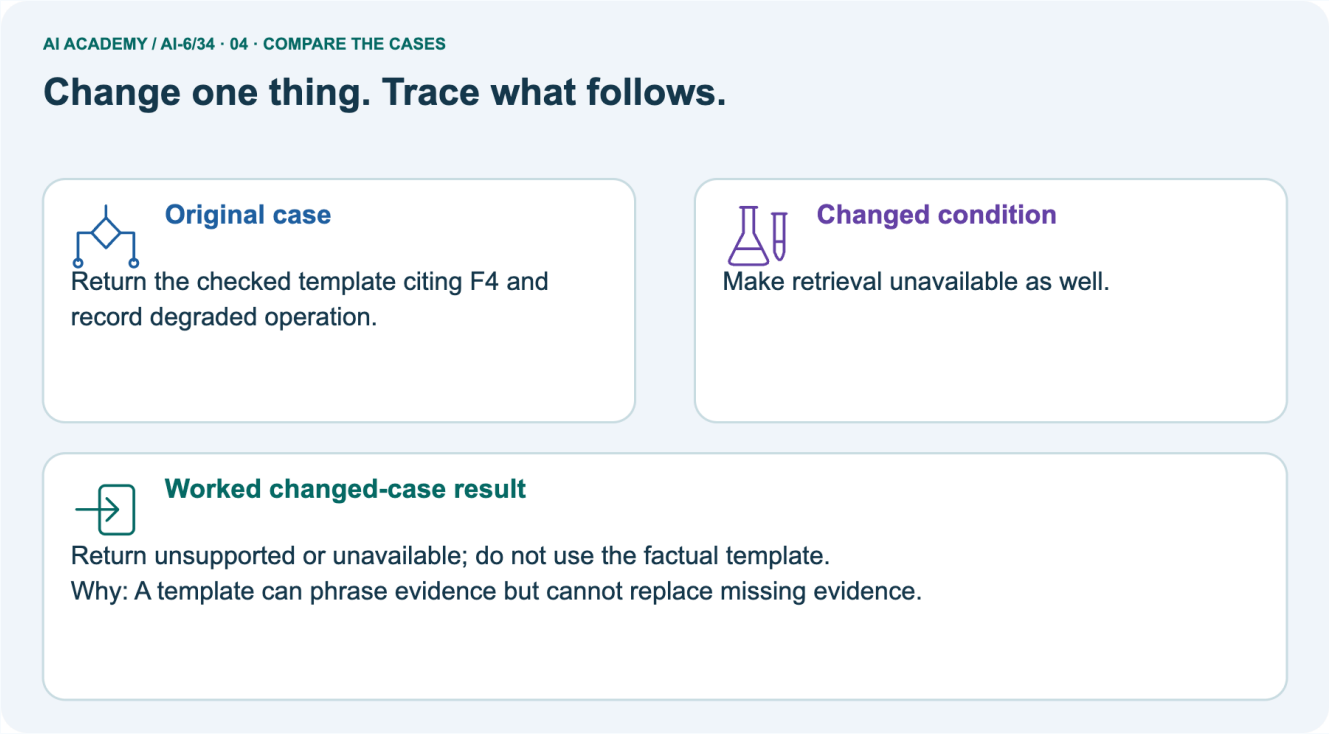


Follow the labelled picture with the matching explanation. Open the vector diagram

Capstone failure testing — **evidence** trace. Four cases: malformed request, student teacher-guide request, missing register, valid current record. One rejects invalid input, one denies access, one reviews and one supports; no restricted reads occur. | Use a **failure matrix** linking condition to expected response and forbidden calls. Test both visible result and operation trace. Preserve failed fixtures for regression checks. | All four meet their expected condition if responses and traces agree. Denial/review are successful contract outcomes, not automatic test failures.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



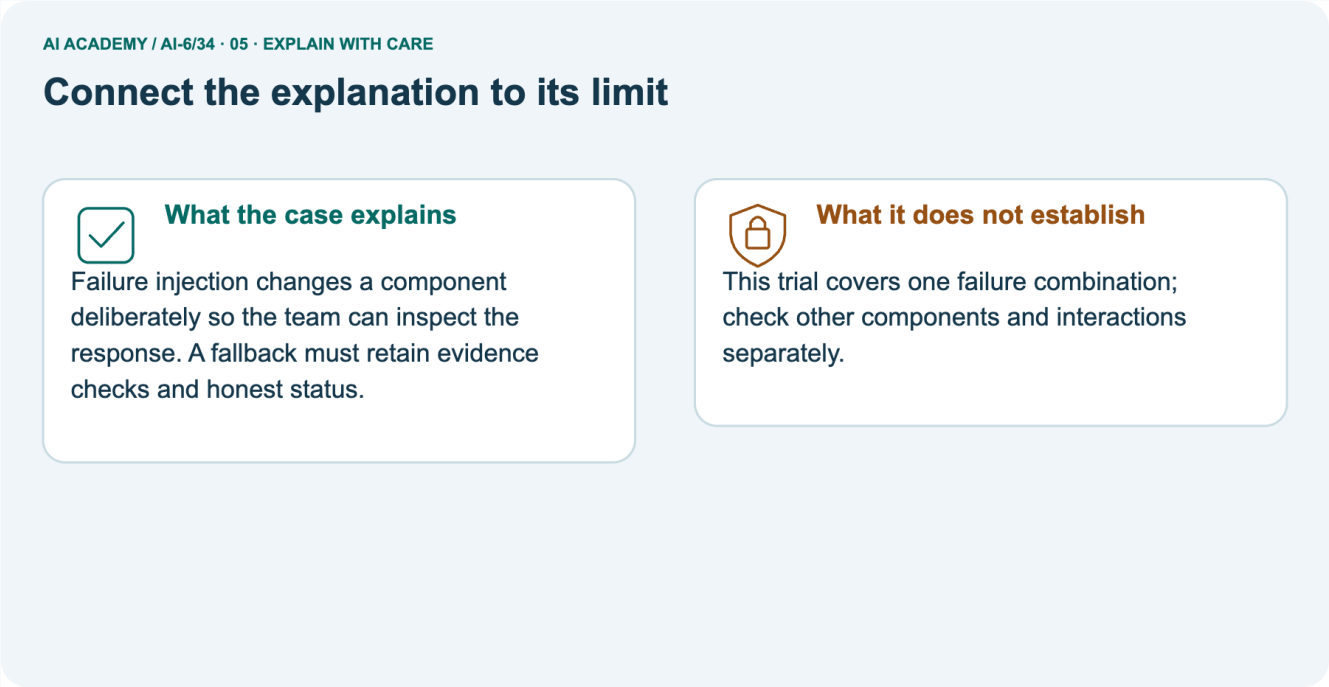
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Return the checked template citing F4 and record degraded operation. Changed condition: Make retrieval unavailable as well. Expected result: Return unsupported or unavailable; do not use the factual template.

Reason: A template can phrase **evidence** but cannot replace missing **evidence**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



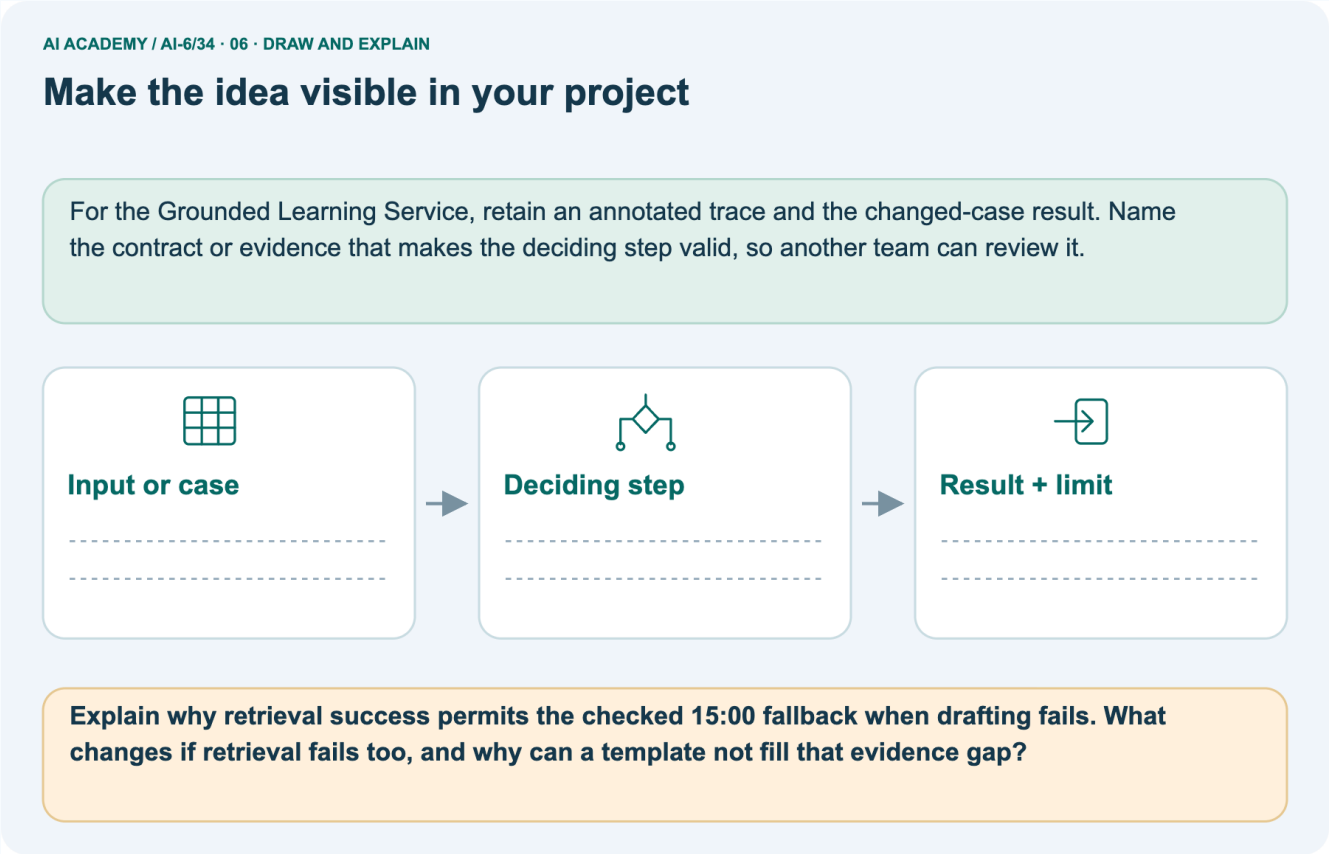
Follow the labelled picture with the matching explanation. Open the vector diagram

Failure injection changes a component deliberately so the team can inspect the response. A fallback must retain **evidence** checks and honest status.

Limit: This trial covers one failure combination; check other components and interactions separately.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Explain why retrieval success permits the checked 15:00 fallback when drafting fails.

What changes if retrieval fails too, and why can a template not fill that **evidence** gap?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Capstone failure testing: the case

02. Follow the deciding steps

03. Capstone failure testing — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.